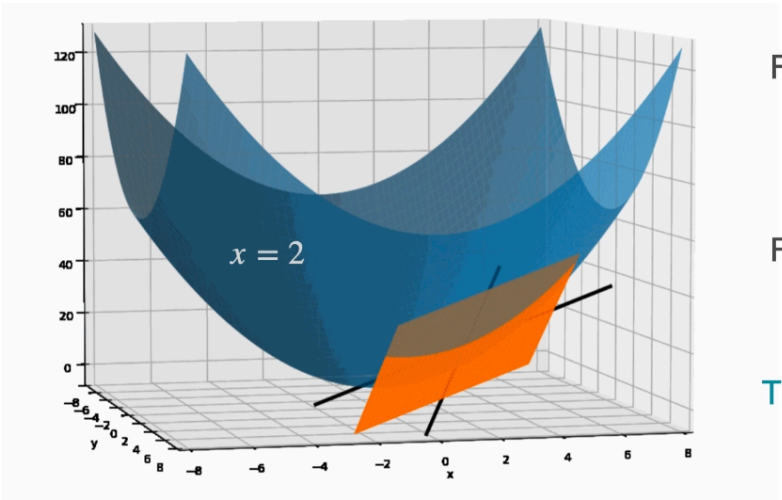
Tangent Plane

The tangent plane contains both tangent lines



Fix y=4 $\quad f(x,4) = x^2 + 4^2$

$$\frac{d}{dx}\big(f(x,4)\big) = 2x$$

Fix x=2 $\quad f(2,y) = 2^2 + y^2$

$$\frac{d}{dy}\big(f(2,y)\big) = 2y$$

The tangent plane contains both tangent lines.